



DEPARTMENT OF  
COMPUTER SCIENCE



# PROGRAMME OBJECTS

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## LEARNING OBJECTIVES

At the end of the lesson, you should be able to:

- Discuss real-world and programme objects.
- Describe the use of variables and methods in programme objects.
- Design structures, unions, and classes.



## WHAT ARE OBJECTS

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What comes to our minds when we think about objects?

Everything in the world?





## OBJECT CHARACTERISTICS

Real-world objects have two main characteristics; they have state and behaviour.

Let us use an example of a dog to understand the characteristics of an object.

OBJECT



STATE:

Name, color, breed, hungry

BEHAVIOUR:

Barking, fetching, wagging  
tail.



## OBJECT OF OBJECTS

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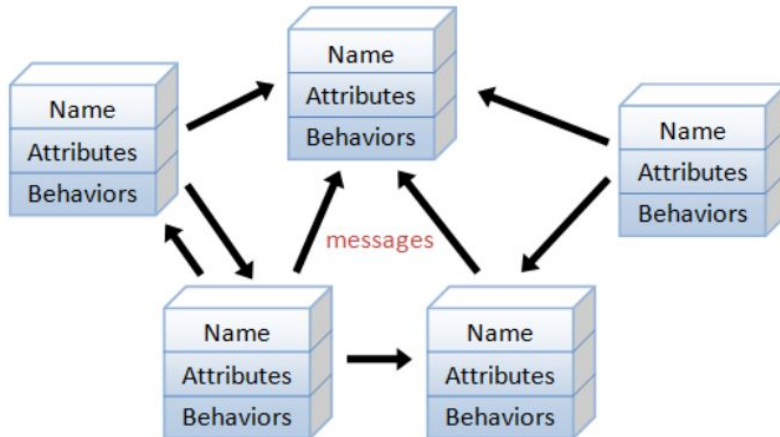
If we spend to some time to think about the real-world objects around us, we will notice the variations in the complexity of real-world objects.

One thing we will also realise is that an object can be part of another object.



## PROGRAMME OBJECTS

The concept of real-world objects is part of what motivates the concept of object-oriented programming (OOP), where programmes are based on objects rather than on separate entities.





## VARIABLES AND METHODS

In programming, an object is defined by its state and behaviour. The state of an object is stored in variables and the behaviour of objects is described by methods.

This means that every object will have their own variables and methods.



## VARIABLES

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Variables are used to store our data in the computer. The variable name is used to retrieve data from the computer.

A program object can have one or more variable associated with it.



## METHODS

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Methods are central to how programme objects work because only the methods of a particular object have access to the variables or state of that object.

Also, different objects can only interact with each other through their own individual methods.



## STRUCTURE

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The structure in C is a user-defined data type that can be used to group items of possibly different types into a single type.

The struct keyword is used to define a structure.

```
struct Person {  
    int age;  
    float heighth;  
    char name[20];  
};
```



## STRUCTURE: ACCESSING AND MODIFYING

To access members of a structure to either assign or modify values, use the dot syntax (.)



## UNION

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Union statements are defined in the same way as structure statements where a user-defined data type that can be used to group items of possibly different types into a single type.

The union keyword is used to define a union

```
union Person {  
    int age;  
    float height;  
    char name[20];  
};
```



## STRUCTURE VS UNION

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A union in 'C' is similar to a structure except that all of its members start at the same location in memory.

A union variable can represent the value of only one of its members at a time.



## CLASS

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Classes are also similar to structures but in addition to having variables, classes also usually have methods or functions.

So, using the dog example, a class will usually consist of variables such as 'name', 'breed' and 'colour'. Also, it will have methods such as a method that returns the name of a dog.



## CLASS

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```
class Dog
{
    // Variables
    char name[20];
    char breed[20];
    char colour[20];

    // Method
    char[] get_name() {
        return name;
    }

    // Method
    void set_name(char[] _name) {
        name = _name;
    }
}
```



## **WHY USE PROGRAMME OBJECTS?**

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Since objects are not dependent on each other, each object can be independently created and maintained.

An object can be used in various parts of our programme.



## WHY USE PROGRAMME OBJECTS?

If an object is not working well or if it is creating problems for other objects in the programme, it can easily be removed and replaced without creating problems for the rest of the code.

This is similar to how a car works!



## WHY USE PROGRAMME OBJECTS?

The fact that the state of an object can only be accessed by the object's methods allows for information hiding.

This means that we can control the amount of information we share about our objects to the outside world.



## SUMMARY

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- Real-world objects have two main characteristics; which are state and behaviour.
- The state of an object is stored in variables and the behaviour of objects is described by methods



## FURTHER READING RESOURCES

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- Bailey, T. (2005). An introduction to the c programming language and software design. July-2005
- Kernighan, B. W., & Ritchie, D. M. (2002). The C programming language.



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THANK  
YOU